

Study of Electroencephalogram Feature Extraction and Classification of Three Tasks of Motor Imagery

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Abstract: Brain-computer interface (BCI) instead of depending on the brain's normal output pathways, can use electroencephalogram (EEG) from the scalp as the representation of brain activity to control external devices. EEG during motor imagery (MI) provides a non-muscular communication way to control external devices and has advantage of non-invasiveness and high time resolution. However the application is still limited by time-consuming training and poor classification rate with multiple categories etc. We recorded 64-channel scalp EEG from eight healthy subjects during imagery tasks of left, right hand movements and stop. EEG was analyzed in time-frequency distribution and spatial topographies were explored too. A one versus one common spatial pattern was applied to construct feature vector and then linear discriminant analysis was used for the classification. For the purpose of real time control in the future, small training size was used and we got discrimination among three types of motor imagery at the accuracy rate about 90%.

Key Words: Brain-computer interface; motor imagery; Electroencephalogram; common spatial pattern

1 INTRODUCTION

At present, the research on the mechanism of human brain and BCI technology has attracted great attention because of its potential application in many fields, especially in rehabilitation and clinical area. Scalp electroencephalogram, which collects brain electrical activity by electrodes attached to the scalp, provides a possible way to translate brain information into control commands for external devices and is widely used in BCI research since its high time resolution and non-invasive characteristics [1-4].

BCI systems aim to help disabled people to interact with the environment, working as a control or manipulation of the movement intent to the external devices [5-7]. Many institutions are engaged in the research of BCI systems based on motor imagery [8-14]. The Wadsworth research center in the United States, the Graz Research Institute in Austria, Germany Freie Universitat Berlin term as well as Tsinghua University in China, have achieved fruitful results [9]. The Wadsworth Center has used EEG rhythms recorded from the scalp over the sensorimotor cortex to control cursor movement in one or two dimensions. Pfurtscheller et al. used the Mu rhythm synchronization of motor imagery to achieve a pointer to one or two dimensional motor control [10]. Graimann et al. used the ERD/ERS phenomenon to classify between the left and right hand motor imagery and applied autonomous walking in virtual reality [11]. Birbaumer laboratory using 0.1-0.2Hz slow cortical potential achieved the selection of the default letters of the late ALS patients [12]. Toyama et al. through ERD/ERS developed AutoNOMOS system achieved brain drive, with recognition accuracy rate at 88% [13]. Gao group completed the robot soccer with three types of motor imagery of left hand, right hand and foot [14].

In the previous research, lots of researchers work in the motor imagery while there still remain some problem, such as relatively low recognition rate in three-class or four-class and large training set needed. This paper has carried out a research of EEG collection and analysis, consisted of EEG acquisition during three categories of motor imagery, feature analysis to extract the motion invention related electrical characteristics, and fast classification for further real-time control of external devices. Thus a direct communication and control channel between a human brain and a computer or other electronic devices could be possible.

2 MATERIALS AND METHODS

2.1 Experimental paradigm

In general, the structure of a typical motor imagery BCI consists of four stages: data acquisition, signal processing, command generation and final application [15]. During experiments, subjects were sitting in a comfortable chair, 1 m from the front of a computer screen. At the beginning of each trial, the screen would appear blank for 2 s. Then fixation indicator "+" presented for 1.5 - 2 s. Then the indicator of either left, or right arrow or stop signal appeared on the screen for 1 - 1.5 s, where the arrows suggested the motor imagery of the left or right hand. Then the sound of "beep" delivered for 1 s, and the subjects were instructed to press different keys to reflect the left, right or stop stimulus presented before. After followed by blank for 1.2 - 1.8 s, the experiment trial was ended. Experimental procedure was shown in Figure 1. One subject took five sessions while each session consisted of 60 trials. Each trial ran about 6.7-8.3 s and randomizations were involved during fixation, stimulus and blank period to avoid the subjects generating prediction.

2.2 Participants

Eight healthy adults (six females and two males with 21 years old on average) took experiments, all right-handed.

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The subjects didn't conduct any related experiments and were instructed the process before the experiments. During the entire experimental procedures, the subjects were informed not to think about anything else except for the imagery of the stimulus of the experiment. Subjects were instructed to relax the muscles, and less blinking, less swallowing and no other acts in the experiment. The subjects were asked to wash their hair before the experiment to prevent excessive oil pollution which will cause the electrode resistance large and thus weak scalp EEG signals will be recorded.

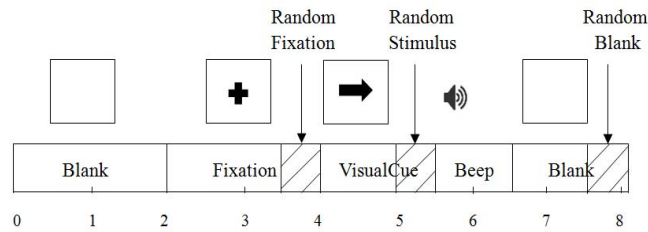


Fig 1. Single trial experimental procedure.

2.3 EEG Recording

EEG was continuously recorded with sampling frequency rate at 1000 Hz by Neuroscan Symamp2 8050, using an EEG cap with 64 silver/silver chloride (Ag-AgCl) electrodes located according to the 10-20 electrode system approved by the International Electrographic Society [15]. The 10-20 system was characterized by the arrangement of the electrodes proportional to the size and shape of the head. The electrode site was in agreement with the anatomy of the cerebral cortex and the electrode name was also consistent with the anatomical partition. Bilateral mastoid electrodes M1 and M2 were reference electrodes. The collected recording potential was the difference between the electrode potential and the potential of the reference electrode [17]. Electrophysiology (EOG) was recorded by the method of bipolar lead. The HEOG electrodes were placed vertically on two sides of eyes, and the VEOG electrodes were placed on the up and down of the left eye. Impedance of all electrodes was maintained below 5 kΩ throughout the experiment [18].

In our study, Curry 7 from Neuroscan was used for EEG collection. E-Prime software from Psychology Software Tools was used to design stimulus presentation and MATLAB 2014 from Mathworks was used for feature analysis and classification.

3 METHODS

3.1 Baseline Correction

The original EEG signals usually contain DC interference thus preprocessing was applied to remove DC components before further data processing. We used the 500 ms data segment before the stimulus for baseline correction. All trials were visually inspected for artifacts using the software Curry 7.

3.2 Wavelet Denoising

The collecting EEG activities are non-stationary stochastic signals with amplitudes in microvolt, easily contaminated with environmental noise and power line. To exact information from EEG signal, denoising is necessary.

The traditional EEG analysis method in frequency domain mainly is power spectral analysis based on the Fourier transform, which assumes that the signal is stationary [19-20]. EEG activity is a typical non-stationary stochastic process. Wavelet transform is a multi-scale signal analysis method with a good time-frequency localization characteristics, which is suitable for the analysis of non-stationary signal transient characteristics [21-24].

In order to remove power line interference and to study EEG activities related with the motor imagery, we used the wavelet transform to denoise the signal and to extract related frequency band.

Assume $x(n)$ to be the EEG signals after the baseline correction, the discrete wavelet transform is defined as:

$$C_{j,k} = 2^{-\frac{j}{2}} \sum_{n=-\infty}^{\infty} x(n) \Psi_{j,k}(2^{-j}n - k) = \langle x(n), \Psi_{j,k} \rangle, j, k \in \mathbb{Z}. \quad (1)$$

where Ψ means the wavelet basis function, j means frequency resolution, and k is the time shift.

Then the Mallat algorithm is used as:

$$x(n) = A_L + \sum_{j=1}^L D_j \quad (2)$$

In the formula, A_L means the approximation coefficient, representing scilicet low frequency component of signal, D_j is the detail coefficient, standing for scilicet high frequency component of signal. L is the number of layers.

In the above formula, the frequency band corresponds to each component is $[0, \frac{f_s}{2^{L+1}}], [\frac{f_s}{2^{L+1}}, \frac{f_s}{2^L}], [\frac{f_s}{2^L}, \frac{f_s}{2^{L-1}}], \dots,$

$[\frac{f_s}{2^2}, \frac{f_s}{2}]$, where f_s is the sampling rate. Since usually

motor imagery related EEG signal concentrated in 8-30Hz, we used the db5 wavelet to decompose EEG signals at six layers. That is,

$$x(n) = a_5 + d_6 + d_5 + d_4 + d_3 + d_2 + d_1 \quad (3)$$

The d5(15.625-31.25Hz) and d6(7.8125-15.625Hz) are used to reconstruct the associated activities with motor imagery.

Raw signal and wavelet reconstruction activities in time domain and the corresponding power spectrum in frequency domain were shown in Figure 2. It is noted that the reconstructed signal after wavelet is smooth and clear, without high frequency interference.

3.3 Common Spatial Pattern (CSP)

During different motor imagery tasks, 64 channels of EEG activities are collected. How to extract motor intention related signal from multiple channels is important. CSP is used to extract features from multi-channel signals as a spatial filtering, which is widely used in two-class classification between two types of tasks. It is based on the

covariance matrix diagonalization. Then in the high dimensional space, there exists the largest variance of the first type and the smallest variance of the second type.

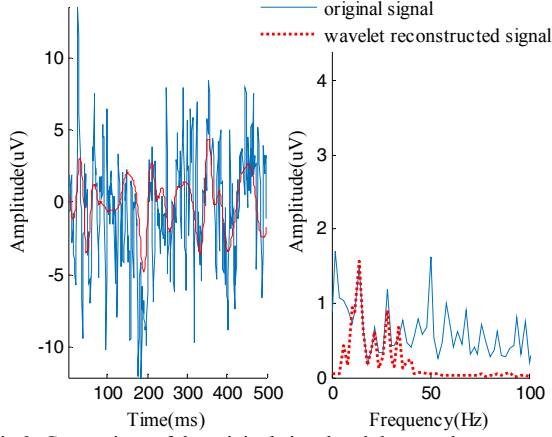


Fig 2. Comparison of the original signal and the wavelet reconstructed signal.

Define the raw EEG signals matrix: $X_{[N \times E]}$, where N is the number of electrodes, E is the sampling points per channel.

The regularized spatial covariance matrix is:

$$C = \frac{XX^T}{\text{trace}(XX^T)} \quad (4)$$

where $\text{trace}(XX^T)$ means the sum of diagonal elements of XX^T .

The average covariance matrix of two classes is calculated, that is $\bar{C}_1$ and $\bar{C}_2$. Define the whole spatial covariance matrix:

$$C = \bar{C}_1 + \bar{C}_2 \quad (5)$$

Make eigendecomposition of C :

$$C = U\Lambda U^T \quad (6)$$

where U is the square matrix whose i^{th} column is the eigenvector of C , and Λ is the diagonal matrix whose diagonal elements are the corresponding eigenvalues.

And the whitening matrix is:

$$P = \Lambda^{-\frac{1}{2}} U^T \quad (7)$$

The P is used to make whitening transformation for the $\bar{C}_1$ and $\bar{C}_2$:

$$S_1 = P\bar{C}_1 P^T, S_2 = P\bar{C}_2 P^T \quad (8)$$

The S_1, S_2 have the same feature vector.

Similarly, make eigendecomposition of S_1, S_2 :

$$S_1 = B\Lambda_1 B^T, S_2 = B\Lambda_2 B^T \quad (9)$$

Especially, $\Lambda_1 + \Lambda_2 = I$.

I is the unit matrix.

When one class has the largest eigenvalues, at the same time the other should have the smallest eigenvalues.

So, we could get the best space filter:

$$W = B^T P \quad (10)$$

Original brain activities can be projected by the filter:

$$Z = WX \quad (11)$$

Finally, the variance of the projected data on each dimension is calculated and standardized:

$$f_p = \log\left(\frac{\text{var}(Z_p)}{\sum_{i=1}^n \text{var}(Z_i)}\right), p = 1 \dots n \quad (12)$$

CSP is an effective method for two-category classification. During the classification among multiple categories, CSP is usually expanded to “one versus one” (OVO) or “one versus rest” (OVR). The OVO is to put multiple types in pairs, and carry out one-over-one feature extraction and classification. OVR is to treat one category as a class, and the rest in multiple categories as the other class. The details of OVO and OVR are introduced in the follows.

3.4 Feature Vector Construction and Pattern Classification

We use CSP based on OVO and OVR respectively. OVO puts EEG signals from three types of motor imagery tasks into “right vs. stop”, “stop vs. left”, “left vs. right” and uses CSP to extract spatial feature respectively. OVR puts the EEG data into “left-right vs. stop”, “stop-right vs. left”, “stop-left vs. right”, then extracting features associated with different imagination of “left”, “right” and “stop”, and finally classifying motion patterns. The procedures of OVO and OVR are shown in Figure 3 and Figure 4.

While applying CSP, the feature matrices are arranged in descending order. That is, the first and last eigenvectors are the best discriminative eigenvectors to classify two types of tasks, and the second and the penultimate feature vectors are the best second discriminative, and so on. After comparing the effect of different number of features, the first three and the last three vectors are selected to construct the feature vectors. After the CSP applied, the feature vectors are sent to the following classifier.

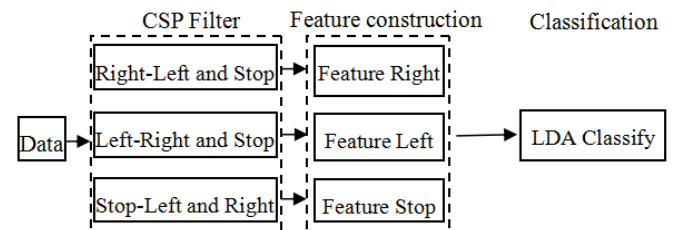


Fig 3. One versus one CSP

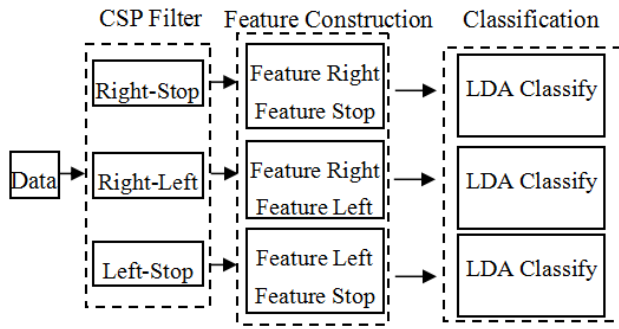


Fig 4. One versus rest CSP

Linear discriminant analysis (LDA) runs fast, operates simply, which is a common algorithm in BCI research for classification [25]. LDA calculates the weight coefficient W and threshold ω_0 of each feature vector of training samples

$$W = (N_1 + N_2)^{-1}(\mu_1 - \mu_2) \quad (13)$$

$$\omega_0 = -\frac{W^T(\mu_1 + \mu_2)}{2} \quad (14)$$

where N_1 , N_2 is the within-class scatter matrix of two types of training samples, μ_1 , μ_2 is the means of feature vector of two types. Then LDA calculates the Fisher distance $g(x)$ to determine the category attribute.

$$g(x) = W^T x + \omega_0 \quad (15)$$

$g(x) \geq 0$, classified as the first type. $g(x) < 0$, classified as the second type.

In order to ensure the independence between the prior information from CSP and the test data for classification, we randomly select half of the data to construct CSP filter. The remaining half of the data is used for classification.

3.5 Cross-validation and Classifier Evaluation

10-fold cross-validation is applied to avoid the contingency of a single verification and the average recognition rate is used to evaluate the classifiers. We divide the remaining half of data into ten parts randomly and send them into the linear classifier. Nine-tenth of that is used to train the classifier, and the other one-tenth is used to test the classification to get one recognition rate. The operation is repeated ten times, and the average classification rate is calculated.

In order to further evaluate the classification performance, we calculate the kappa score. The kappa value corrects false consistency generated by random factor from K-fold cross-validation. It is defined as

$$k = \frac{\Pr(a) - \Pr(e)}{1 - \Pr(e)} \quad (16)$$

where $\Pr(a)$ means the calculated consistency accuracy and $\Pr(e)$ is the false consistency by random factor.

The kappa score ranges between -1 and 1, where -1 means the worst classification and 1 indicates the best. When $\text{Kappa} > 0.75$, the consistency is high, and the classification is good. When $0.4 < \text{Kappa} < 0.75$, the consistency and

classification is of general, When $\text{Kappa} < 0.4$, the consistency is poor.

In order to evaluate classification based on CSP more comprehensively, the Receiver Operating Characteristic (ROC) curve and the Area Under Curve (AUC) value is under study. ROC uses true positive rate (TRP) and false positive rate (FRP) to indicate the accuracy of the classification method graphically. The closer the curve is to the upper left corner, the better the classifier is. Concepts related to ROC are defined as follows.

$$\text{TRP} = \frac{TP}{TP + FN}, \text{FRP} = \frac{FP}{FP + TN} \quad (17)$$

Where True Positive (TP) is the number of positive samples classified as positive, False Negative (FN) is the number of positive samples classified as negative, False Positive (FP) is the number of negative samples classified as positive and True Negative (TN) is the number of negative samples classified as negative.

The AUC value is the area covered by the ROC curve. Obviously, the larger the AUC is, the better the classifier is. The relationship between AUC and classifier is as follows. $\text{AUC} = 1$, the perfect classifier. When $\text{AUC} = 0.5$, the classifier is the same as random guess and the model has no classification value. When $0.5 < \text{AUC} < 1$, a classifier is better than random guess. When $\text{AUC} < 0.5$, a classifier is worse than random guess.

4 RESULT

In order to extract feature related to motion intention among multiple cortical areas, the spatial distributions of EEG signals are explored by topographic maps as Figure 5. [26]. In the figure, top two rows are during motor imagery of left (L), the middle two rows during right (R) and the bottom two rows during stop (S). Colors changing from red to blue indicate power amplitude from high to low. The event-related desynchronization (ERD) indicated by color in blue and event-related synchronization (ERS) indicated by color in red show that the energy could be used as the features for classification. During the unilateral motor imagery, the μ rhythm (8-12Hz) and β rhythm (14-30Hz) on the contralateral brain area appear a significant amplitude increase (synchronization), while that of the ipsilateral brain area appear an amplitude decrease (desynchronization).

In order to explore temporal and frequency characteristics of EEG during the motor imagery, the time-frequency distributions for one subject by Hilbert-Huang transform are illustrated in Figure 6. Colors changing from red to blue indicate power change. Figure 6 shows the different time-frequency distributions from channel C3 for motor imagery of left, right hand and stop. Different patterns are found during different motor imagery.

The discrimination of different types of motor imagery is based on feature vector constructed by "one versus one" OVO-CSP or "one versus rest" OVR-CSP and then classified by LDA. The recognition rate based on OVO-CSP and OVR-CSP are shown in the table I. From the table, the OVO-CSP reaches about 90% recognition rate for

eight subjects on average and has a better discriminative ability than the OVR-CSP. 10-fold cross-validation is used to avoid over-fitting. When extracting features by OVO-CSP, we select the first three and the last three eigenvectors of the pre-processed data.

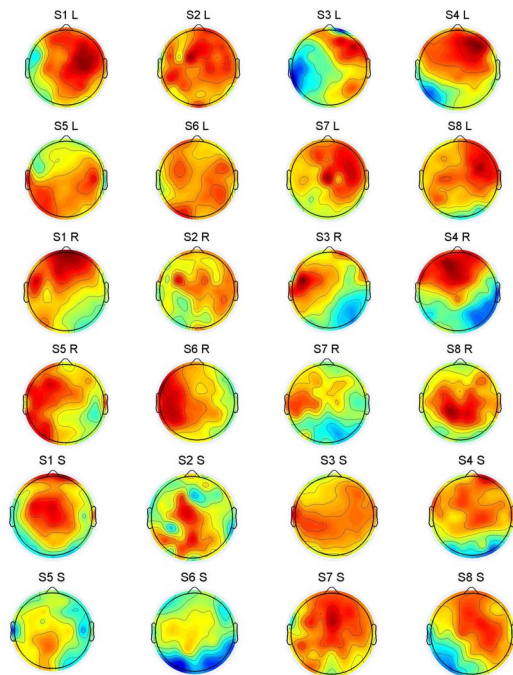


Fig. 5. Topographic maps with 8-30 Hz frequency band during motor imagery of left, right and stop for eight subjects.

Table I. Average Recognition Rate Based on OVO-CSP and OVR-CSP

Subject	1	2	3	4	5	6	7	8
OVO	89.9%	90.1%	89.0%	90.4%	90.3%	91.8%	90.0%	87.8%
OVR	71.3%	73.3%	71.7%	70.0%	77.8%	77.8%	74.0%	74.2%

To evaluate the performance of classifiers, the ROC curves of three OVO-CSP classifications are shown in Figure 7. In the ROC curve, the vertical axis means the true positive rate, and the horizontal axis indicates the false positive rate. And the closer the curve is to the upper left corner, the better the consistency and accuracy is. As Figure 8 indicates, the curves of three OVO classification almost reach the upper left corner, and the AUC indicated by the area under ROC curve above 0.85, which indicates a good effort of classification.

Recognition rate reveals the overall recognition rate. What is more, kappa score could correct the false consistency introduced by random factors in K-fold cross-validation. The kappa scores of the OVO classification are shown in the table II. Kappa values are around 0.8, which indicates that the difference of K-fold cross-validation is small. In other words, the false consistency introduced by cross-validation is small and a good identification is achieved.

Considering there are only 20 trials per category in one session, it is indicated that by using the small training sample motor imagery classification is applicable

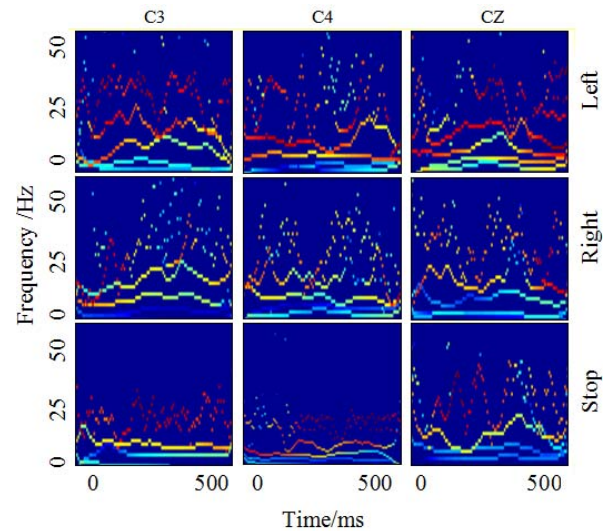


Fig. 6. Time-frequency distributions of channel C3, C4 and CZ from subject#1 of motor imagery of left right and stop

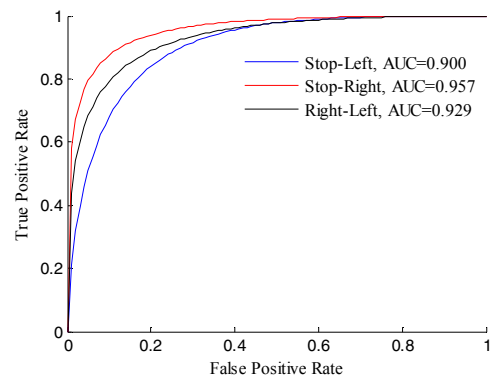


Fig. 7. ROC curves of three OVO CSP classifications

Table II. Kappa Scores of OVO Classification

Subject	1	2	3	4	5	6	7	8
kappa	0.80	0.80	0.78	0.80	0.81	0.84	0.80	0.76

5 CONCLUSION

In the paper, by exploring features in time-frequency and topographies, the signals are quantitatively analyzed by wavelet decomposition and feature vectors are constructed by one versus one CSP. The classification results based on LDA show a recognition rate around 90% for eight subjects on average. An acceptable recognition is the pre-requisite for the future BCI applications based on motor imagery.

6 DISCUSSION

Generally, EEG signals are non-stationary stochastic process. Electrical activities of motor brain area are related to motor imagery and the electrical signals vary obviously. During different types of motor imagery, different brain areas are involved in different degrees. The location of the electrode with most discriminative information varies from individual to individual. How to extract the discriminative motion related features across multiple channels by spatial filter is under study.

In practical application, the acquisition of high-density lead EEG signals is a time-consuming and laborious task, which is inconvenient for the further practice of BCI. In theory, the EEG signals collected from electrodes on motor cortex reveal motion intention. During motor imagery, there exists variability among different subjects and even different trials for the same subject. Different subjects respond differently to the motor imagery. In addition, for the same subject, the recognition rate calculated from one session is different from other sessions. Assuming that during one session, subject's physiological state of the brain is similar. When one session is over, the subject will have a rest and relax, preparing for the next session. During this process, subject's physiological state of the brain would change and thus that of next session would be different from the first. That is, in the EEG analysis, we should not only think about the differences among subjects but also the differences among sessions of the same subject. How to generalize the recognition across different subjects or different days for the same subject is one of the key issues for application of BCI systems in the future.

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